

CVE-2025-40780: Cache poisoning due to weak PRNG

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Content

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CVE: [CVE-2025-40780](#)

Title: Cache poisoning due to weak PRNG

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Posting date: 22 October 2025

Program impacted: BIND 9

Versions affected:

BIND

- 9.16.0 -> 9.16.50
- 9.18.0 -> 9.18.39
- 9.20.0 -> 9.20.13
- 9.21.0 -> 9.21.12

BIND Supported Preview Edition

- 9.16.8-S1 -> 9.16.50-S1
- 9.18.11-S1 -> 9.18.39-S1
- 9.20.9-S1 -> 9.20.13-S1

(Versions prior to 9.11.0 were not assessed.)

Severity: High

Exploitable: Remotely

Description:

In specific circumstances, due to a weakness in the Pseudo Random Number Generator (PRNG) that is used, it is possible for an attacker to predict the source port and query ID that BIND will use.

Impact:

BIND can be tricked into caching attacker responses, if the spoofing is successful.

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Authoritative services are believed to be unaffected by this vulnerability but it is important to read: <https://kb.isc.org/docs/why-does-my-authoritative-server-make-recursive-queries>

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Resolvers are affected by this vulnerability.

CVSS Score: 8.6

CVSS Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:N/I:H/A:N

For more information on the Common Vulnerability Scoring System and to obtain your specific environmental score please visit: <https://nvd.nist.gov/vuln-metrics/cvss/v3-calculator?vector=AV:N/AC:L/PR:N/UI:N/S:C/C:N/I:H/A:N&version=3.1>.

Workarounds:

No workarounds known.

Active exploits:

We are not aware of any active exploits.

Solution:

Upgrade to the patched release most closely related to your current version of BIND 9:

- 9.18.41
- 9.20.15
- 9.21.14

BIND Supported Preview Edition is a special feature preview branch of BIND provided to eligible ISC support customers.

- 9.18.41-S1
- 9.20.15-S1

Acknowledgments:

ISC would like to thank Prof. Amit Klein and Omer Ben Simhon from Hebrew University of Jerusalem for bringing this vulnerability to our attention.

Document revision history:

- 1.0 Early Notification, 8 October 2025
- 1.1 Revised public disclosure date, 14 October 2025
- 1.2 Revised the list of fixed versions, 15 October 2025
- 2.0 Public disclosure, 22 October 2025

Related documents:

See our [BIND 9 Security Vulnerability Matrix](#) for a complete listing of security vulnerabilities and versions affected.

Do you still have questions? Questions regarding this advisory should be mailed to bind-security@isc.org or posted as confidential GitLab issues at [[https://gitlab.isc.org/isc-projects/bind9/-/issues/new?issue\[confidential\]=true](https://gitlab.isc.org/isc-projects/bind9/-/issues/new?issue[confidential]=true)](<https://gitlab.isc.org/isc-projects/bind9/-/issues/new?issue%5Bconfidential%5D=true>).

Note:

ISC patches only currently supported versions. When possible we indicate EOL versions affected. For current information on which versions are actively supported, please see <https://www.isc.org/download/>.

ISC Security Vulnerability Disclosure Policy:

Details of our current security advisory policy and practice can be found in the ISC Software Defect and Security Vulnerability Disclosure Policy at <https://kb.isc.org/docs/aa-00861>.

The Knowledgebase article <https://kb.isc.org/docs/cve-2025-40780> is the complete and official security advisory document.

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